Evidence and Sources

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Evidence and Sources before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Prompt Experiments

Without opening the old lesson, explain one key idea from Prompt Experiments. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Image and Voice Pipelines

Without opening the old lesson, explain one key idea from Image and Voice Pipelines. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Recommendation Scores

Without opening the old lesson, explain one key idea from Recommendation Scores. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Evidence and Sources trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Claim** support and citation evaluation. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Evidence and Sources explanation against the supplied facts before discussing it.

Use the worked case for **Claim** support and citation evaluation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Five **claim**s; four cited; three supported by cited passages; one cited ID is nonexistent. Compute support and citation presence and explain link limits.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Evidence and Sources target to a trace and a justified conclusion.

Two catalogue records list 96 and 112 pages for a title. A helper averages them and reports 104. Explain the mistake and a better verification **process**.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Evidence and Sources case and show the deciding evidence.

An answer says 80 pages, but the catalogue for that edition says 120.

What **evidence** guides correction?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

An answer cites a page about opening hours to support a **claim** about book prices. Is a citation enough? What must be checked?

My **independent** answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Evidence and Sources adds to Transparent Book Recommender and keep the evidence.

Separate profile-based ranking **evidence** from factual book information and cite the latter.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Two catalogue records list 96 and 112 pages for a title. A helper averages them and reports 104. Explain the mistake and a better verification **process**.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Evidence and Sources facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace the worked case

> Trace the worked case. Apply the supplied Evidence and Sources facts and keep your reasoning.

An answer has four factual **claim**s. Three have citations, but only two are supported by their cited passages. All links exist. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Evidence and Sources facts and keep your reasoning.

A report omits this qualification: Citation counts do not establish truth beyond the supplied **source**s or the correctness of uncited **claim**s.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Evidence and Sources facts and keep your reasoning.

Five **claim**s; four cited; three supported by cited passages; one cited ID is nonexistent. Compute support and citation presence and explain link limits.

> **My reasoning and evidence**

WORKBOOK · 16 OF 21

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Evidence and Sources facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Evidence and Sources

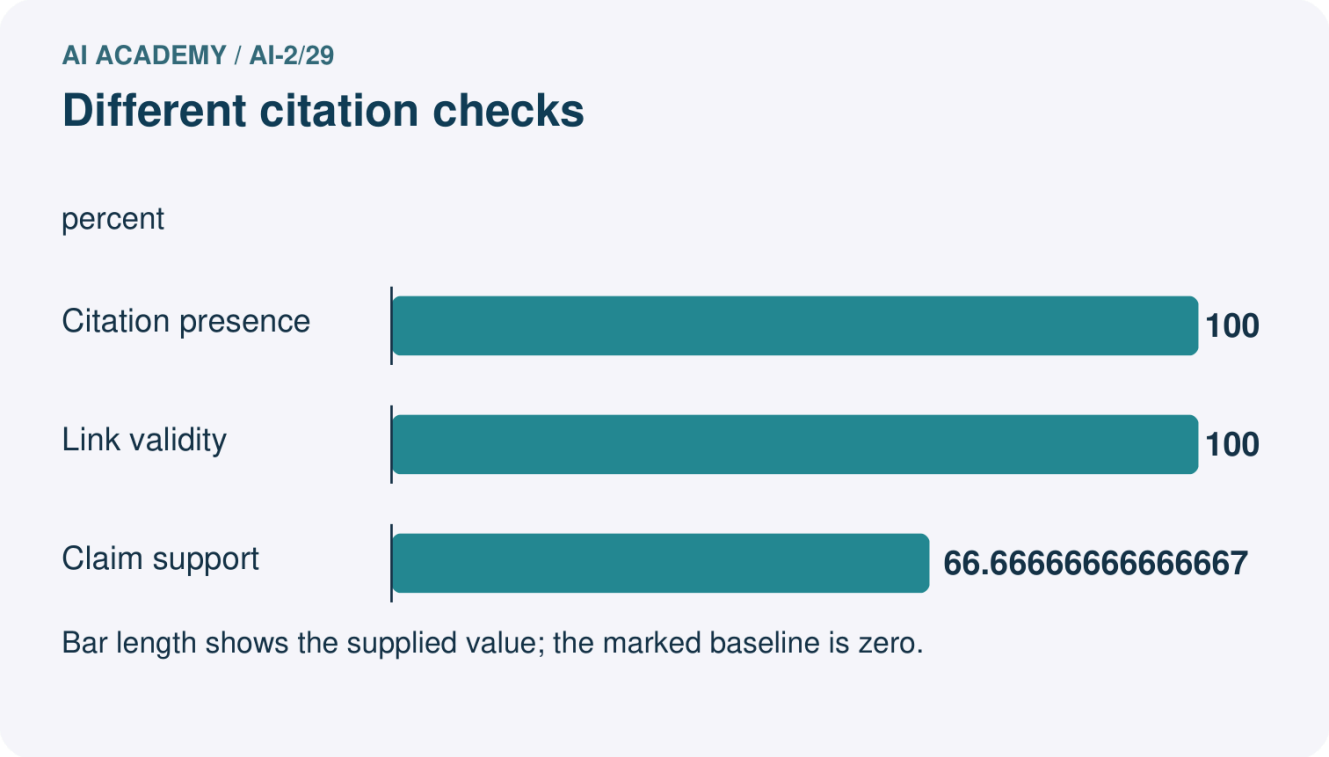
> A fictional catalogue answer contains three factual statements and cites existing records.

> Explain the labelled step before checking the changed condition.

A fictional catalogue answer contains three factual statements and cites existing records.

Record R: book Ocean Maps has 80 pages and is available on shelf 2.

Claims: 80 pages [R]; shelf 2 [R]; suitable for every child [R]. All three citations resolve to R.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/29

Evidence and Sources

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Break into claims

Count three factual claims rather than treating the answer as one blob.



2. Check links

All three citation links resolve to the intended R.



3. Check exact source support

R supports page count and shelf, not universal suitability.



4. Report separately

Citation presence 3/3; link validity 3/3; supported claims 2/3.

CHANGE ONE THING

Change the third claim to "the record does not assess suitability".

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why is the original answer not fully supported despite having citations on every **claim**?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Attach **source** IDs to recommender explanation facts and separately audit whether each **source** supports its precise statement.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Attach **source** IDs to recommender explanation facts and separately audit whether each **source** supports its precise statement.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved **independent** assessment for a separate first attempt before teacher feedback.

A fictional catalogue answer contains three factual statements and cites existing records.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Citation presence 3/3; **link validity** 3/3; supported **claim**s 2/3.

> 1. Check exact **source** support

2. Check links

3. Report separately

4. Break into **claim**s

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report separately Citation presence 3/3; **link validity** 3/3; supported **claim**s 2/3. An existing citation is not automatically **evidence** for every nearby sentence.

Read the cited passage and compare its exact scope with the **claim**. The universal suitability statement extends beyond what R says.

Remove or qualify unsupported content; adding more copies of the same link does not create the missing **evidence**.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change the third **claim** to “the record does not assess suitability”. Which conclusion is supported by the supplied facts?

1. That statement accurately describes the supplied record’s limits.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: That statement accurately describes the supplied record’s limits. R gives pages and shelf only; it contains no suitability assessment.

Repair a missing building block

Review Prompt Experiments (ai-2/28). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: Name the case P passes and Q fails, and explain why it matters..

Return to this week and explain what you changed.

Review Image and Voice Pipelines (ai-2/26). Goal: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

Short practice: Which stage first fails, and why should the faulty transcript not be silently overwritten?. Return to this week and explain what you changed.

Review Recommendation Scores (ai-2/23). Goal: Compute a weighted score. Short practice: Why does B not become the winner when its score is largest?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

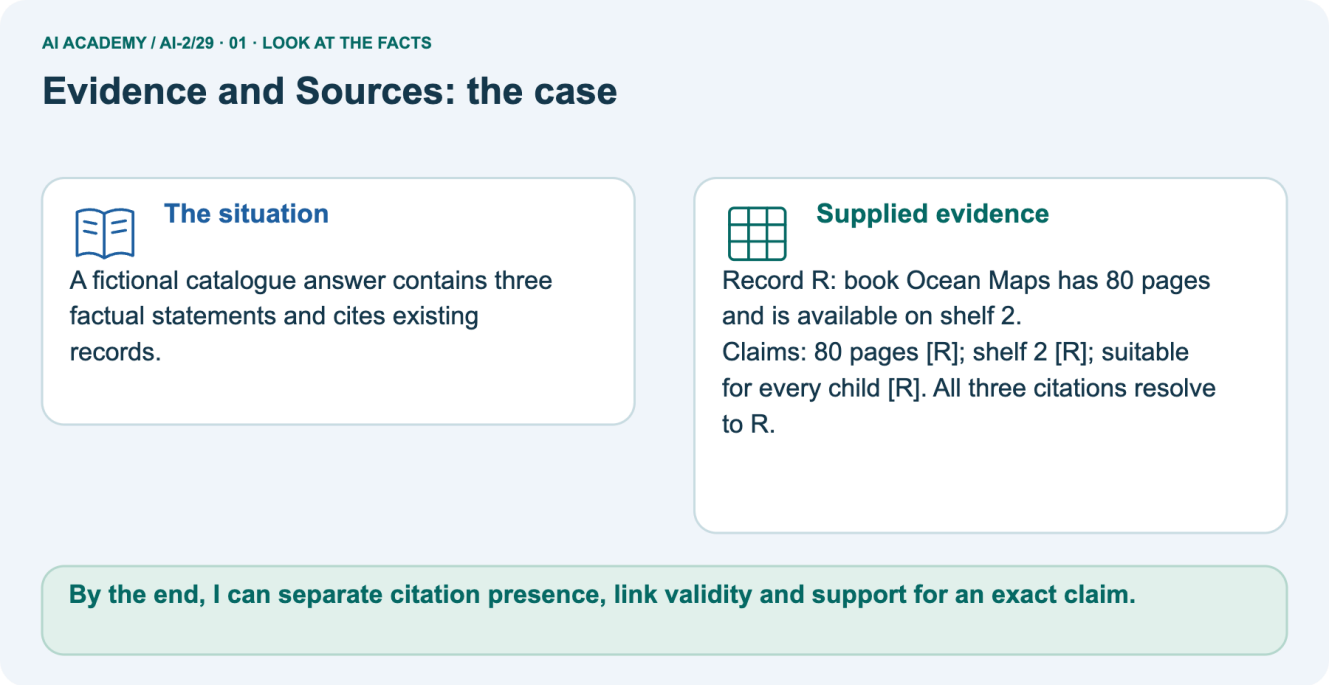
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Evidence and Sources picture from its labels, then record a relevant limit.

01. Evidence and Sources: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

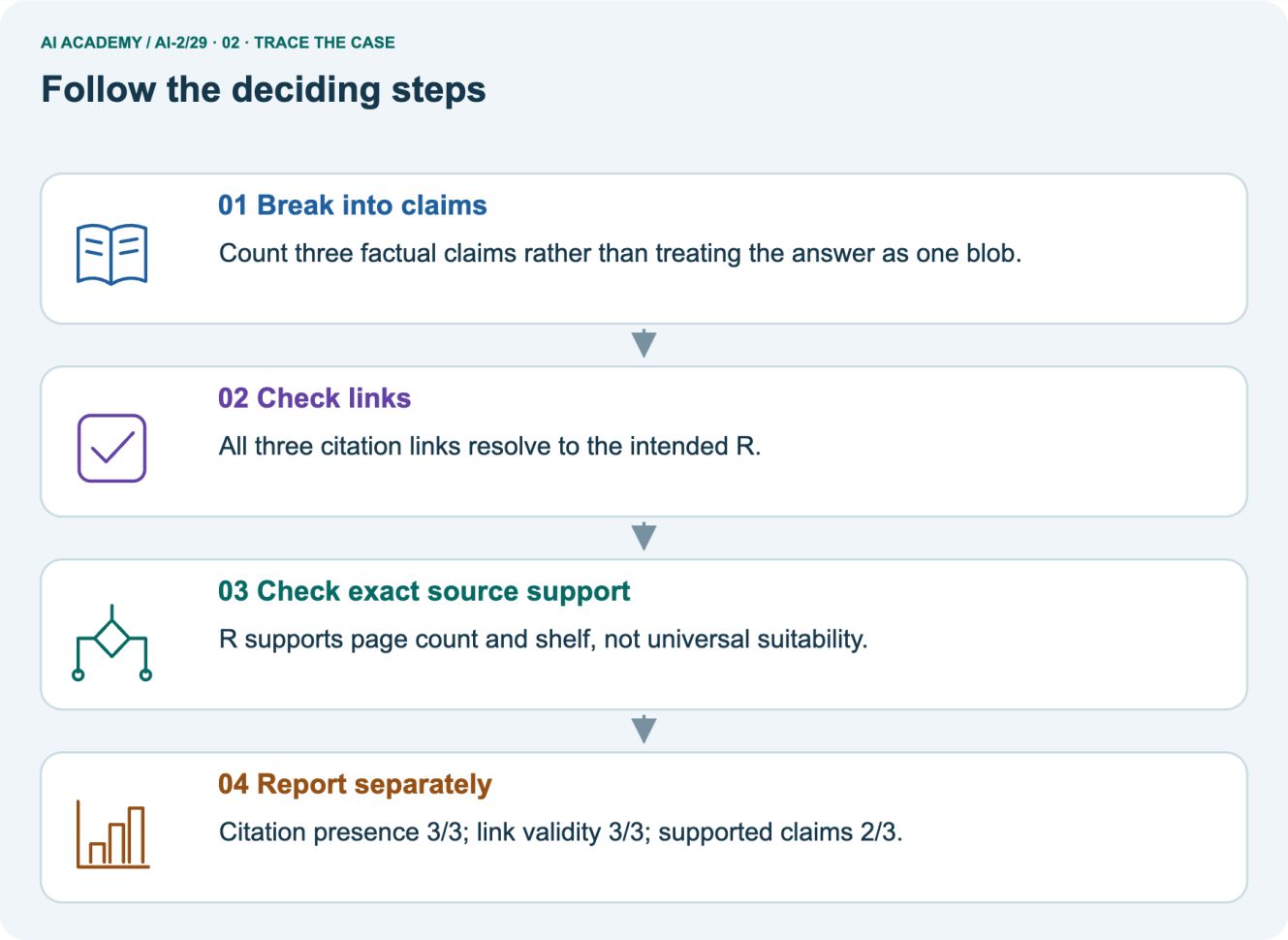
A fictional catalogue answer contains three factual statements and cites existing records. Supplied facts: Record R: book Ocean Maps has 80 pages and is available on shelf 2.

Claims: 80 pages [R]; shelf 2 [R]; suitable for every child [R]. All three citations resolve to R.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



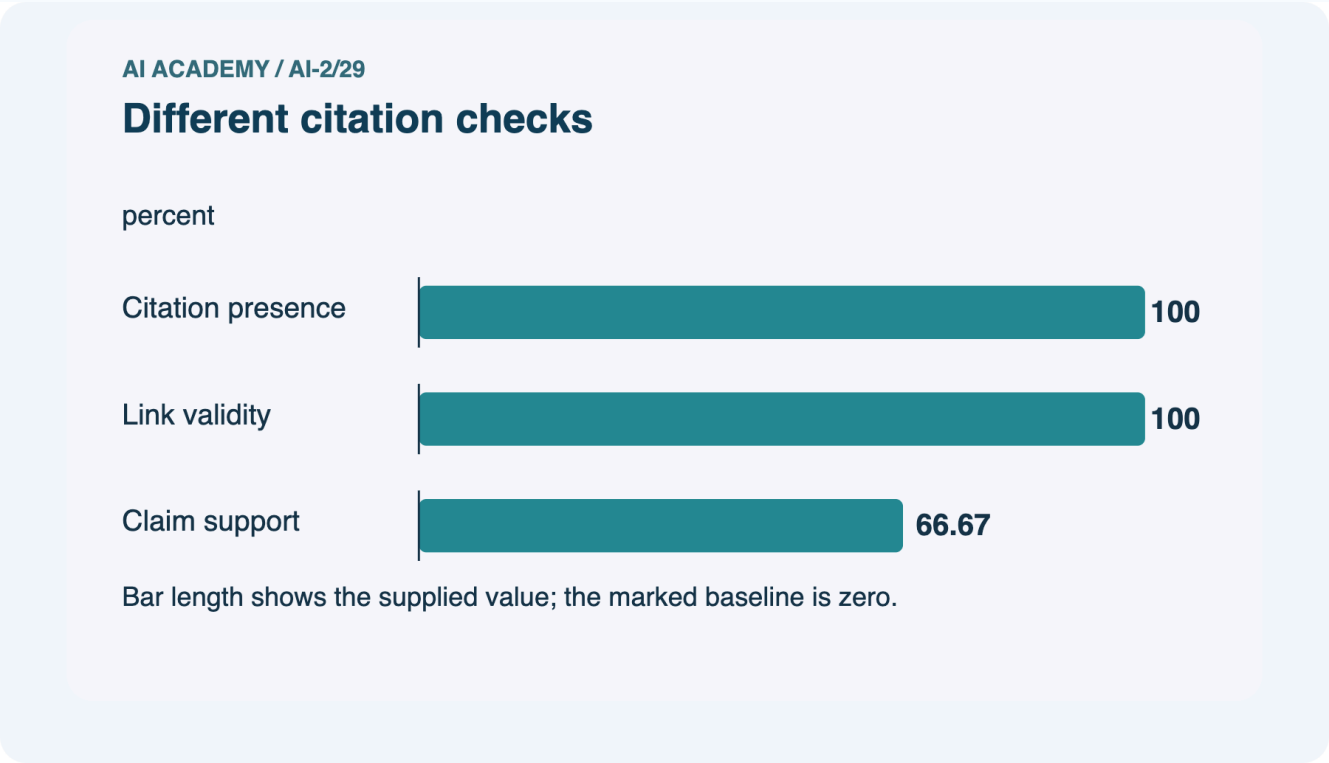
Follow the labelled picture with the matching explanation. Open the vector diagram

Break into **claim**s: Count three factual **claim**s rather than treating the answer as one blob. → Check links: All three citation links resolve to the intended R. → Check exact **source** support: R supports page count and shelf, not universal suitability. → Report separately: Citation presence 3/3; **link validity** 3/3; supported **claim**s 2/3.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Different citation checks



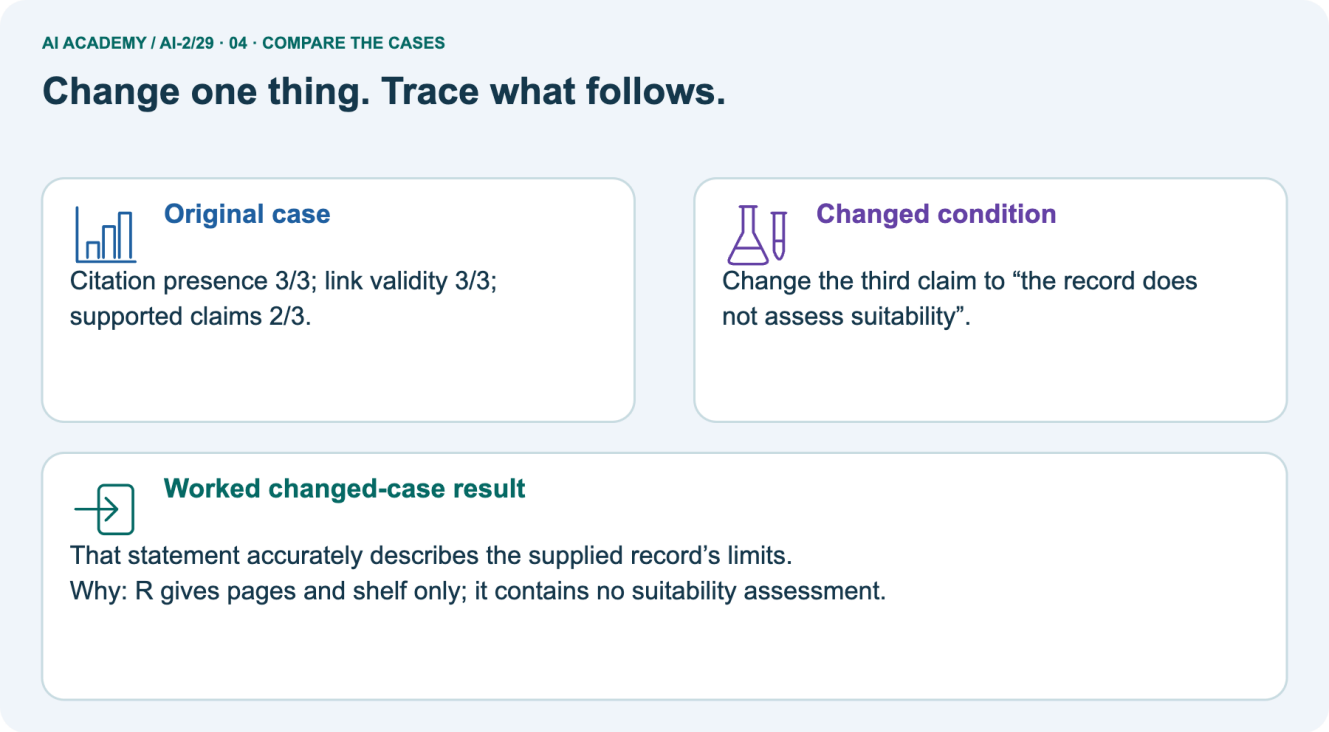
Follow the labelled picture with the matching explanation. Open the vector diagram

Different citation checks. Units: percent. Citation presence: 100. **Link validity**: 100. **Claim** support: 66.66666666666667.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

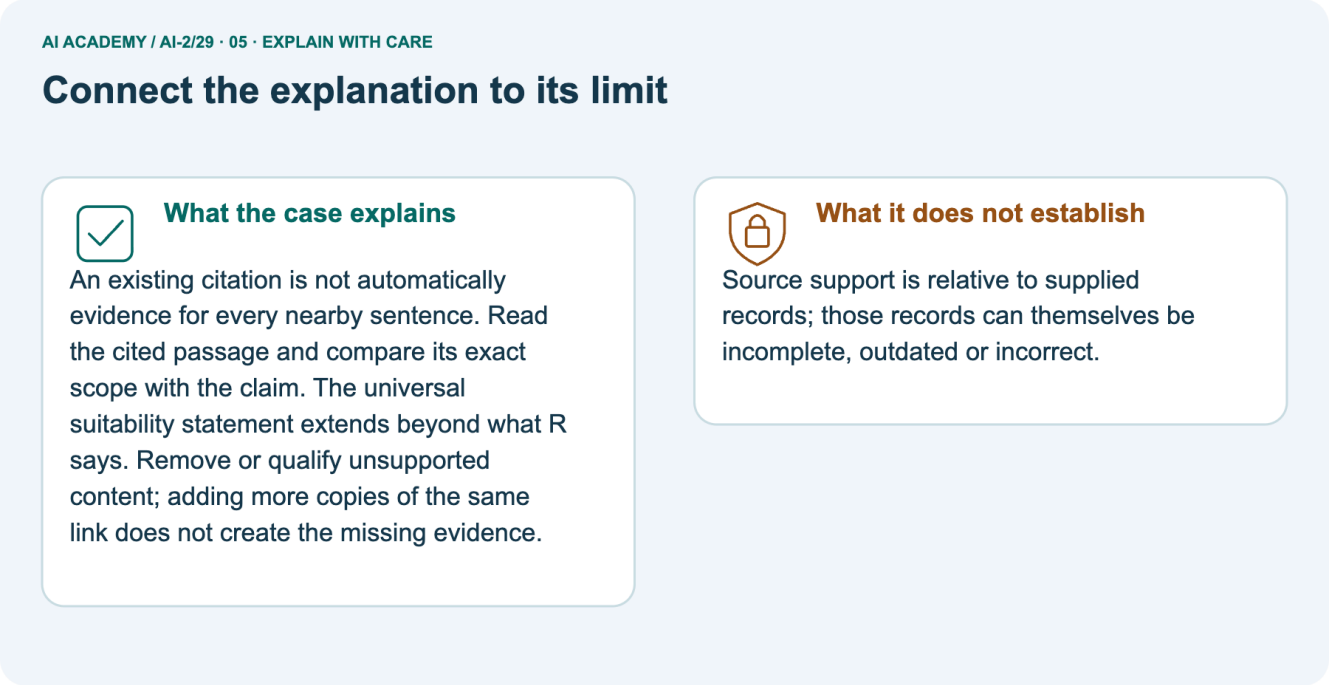
Original case: Citation presence 3/3; **link validity** 3/3; supported **claim**s 2/3. Changed condition: Change the third **claim** to “the record does not assess suitability”.

Expected result: That statement accurately describes the supplied record’s limits. Reason: R gives pages and shelf only; it contains no suitability assessment.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An existing citation is not automatically **evidence** for every nearby sentence. Read the cited passage and compare its exact scope with the **claim**. The universal suitability statement extends beyond what R says.

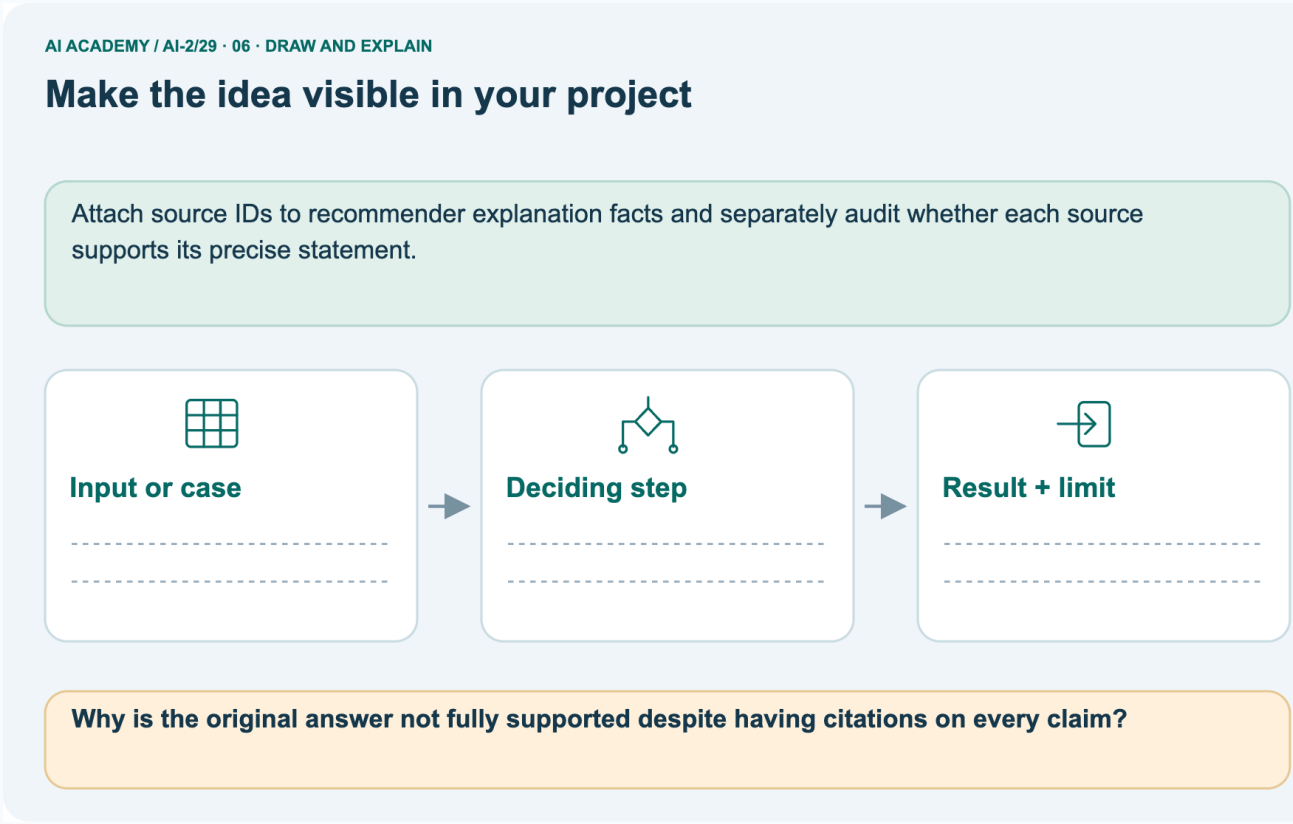
Remove or qualify unsupported content; adding more copies of the same link does not create the missing **evidence**.

Limit: **Source** support is relative to supplied records; those records can themselves be incomplete, outdated or incorrect.

> Explain the case, then point to a limit on the **claim**.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Attach **source** IDs to recommender explanation facts and separately audit whether each **source** supports its precise statement. Draw your **input**, deciding step, result and limit.

Supported question: Why is the original answer not fully supported despite having citations on every **claim**?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.